

A Methodology to Predict The Side Effects of Newly Developed Drugs Using Docking Score From Previously Approved Drugs and Machine Learning Techniques

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Through the establishment of pharmacology networks of drugs interacting with many proteins target can help us to explore into drug re-purposing, side effect prediction, the development of more efficacious drugs and other useful things. Linking known and potential drugs to drug targets and off targets can give us important information which can be used as a feature information for advanced machine learning algorithms which in turn can generate information for newly developed drugs eventually will improve the throughput in drug development process and help develop safer drugs. In this work we proposed a method of using molecular docking score and a virtual screening panel to predict the adverse drug reactions for newly developed drugs. In order to test the hypothesis that molecular docking score can be used to predict the side effects using a minimal panel of proteins, we have performed the following steps. We first established a virtual safety panel repository for well known proteins. Then link potential risk associated with the panel to known adverse reactions. Established a correlation between the generated docking scores and the presence of adverse impact linking the panel of proteins to approved drugs. Finally used the

docking score to learn and train advanced machine learning algorithms in order to use the learned model to predict adverse effects for newly developed drugs. In order to get more accurate result we used high performance computing to facilitate large number of docking calculations on large computer clusters and also extracted an ensemble of potentially druggable conformations through simulation of protein motion.

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